# Detecting Machine-Generated isiZulu Text

## Fine-Tuning AfroXLMR-Large-29L for Binary AI-Text Classification

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**Abstract**

This report describes the fine-tuning of AfroXLMR-Large-29L — an XLM-RoBERTa Large model pre-trained on African language corpora — to classify isiZulu text as either human-authored or machine-generated. Training used a balanced dataset of 16,377 labelled samples over 4 epochs with early stopping. The best checkpoint (epoch 2) achieves 99.40% F1 on the validation set and 99.26% F1 on a held-out test set of 2,022 samples, with a ROC-AUC of 99.97% and an MCC of 0.9852. A five-part data integrity audit confirms the result reflects genuine generalisation. The train-to-test F1 degradation is 0.54 percentage points, consistent with a model that has learned stable linguistic patterns rather than memorised training examples.

## 1. Dataset

The dataset consists of isiZulu passages labelled as human-authored (label 0) or machine-generated (label 1). Machine texts were produced by translating human passages to English and back to isiZulu via a neural machine translation pipeline. This preserves topical content while altering linguistic texture — vocabulary choice, morphological variety, and idiomatic phrasing — which is the signal the classifier must learn.

| Split | Total | Human (label 0) | Machine (label 1) | Balance |
|---|---|---|---|---|
| Train | 16,377 | 8,207 (50.1%) | 8,170 (49.9%) | 50 / 50 |
| Eval | 1,991 | 994 (49.9%) | 997 (50.1%) | 50 / 50 |
| Test | 2,022 | 1,013 (50.1%) | 1,009 (49.9%) | 50 / 50 |

Table 1. Dataset split composition.

All three splits are near-perfectly balanced, so accuracy and macro-F1 are interchangeable and no class-weighting is needed. Human texts average 1,492 characters per passage (median 1,534); machine texts average 1,432 characters (median 1,549). Mann-Whitney U tests confirm no significant length shift between splits ($p > 0.52$ for all three cross-split comparisons), establishing that train, eval, and test are drawn from the same underlying distribution.

## 2. Model and Training

### 2.1 Base model

AfroXLMR-Large-29L is an XLM-RoBERTa Large encoder further pre-trained on 17 African languages including isiZulu. Compared to standard XLM-RoBERTa Large, it produces substantially better sub-word representations for Bantu agglutinative morphology. The architecture is: 24 transformer layers, 16 attention heads, hidden size 1,024, FFN intermediate size 4,096, vocabulary of 250,002 tokens, approximately 560 M parameters.

### 2.2 Classification head and fine-tuning setup

A two-class linear head (dense → GELU → dropout 0.1 → linear projection to 2 logits) was attached to the [CLS] token representation and initialised randomly. The full model was then trained end-to-end. Texts were tokenised to a maximum of 512 sub-word tokens.

| Hyperparameter | Value |
|---|---|
| Per-device batch size | 8 (gradient accumulation × 4 → effective 32) |
| Learning rate | 2 × 10■■ (cosine schedule, 10% warmup) |
| Weight decay | 0.01 |
| Mixed precision | fp16 |
| Max epochs / patience | 5 epochs / early stopping patience 2 |
| Model selection metric | Eval F1 (binary) |
| Hardware / time | NVIDIA RTX 5090 (33.7 GB) ≈ 22 minutes |
| Random seed | 42 |

Table 2. Training hyperparameters.

### 2.3 Training dynamics

Training stopped after epoch 4 when early stopping triggered — neither epoch 3 nor epoch 4 exceeded the epoch 2 eval F1. The best checkpoint was saved at epoch 2.

| Epoch | Eval F1 | Eval Acc | Eval Loss | Train F1 | Gap | Note |
|---|---|---|---|---|---|---|
| 1 | 98.42% | 98.39% | 0.0951 | 98.80% | +0.38% | |
| 2 | 99.40% | 99.40% | 0.0324 | 99.80% | +0.40% | Best — saved |
| 3 | 97.22% | 97.14% | 0.1868 | 98.50% | +1.28% | |
| 4 | 99.15% | 99.15% | 0.0698 | 99.85% | +0.70% | Stop triggered |

Table 3. Per-epoch results. Gap = train F1 minus eval F1 (train F1 measured on a stratified 2,000-sample subset of the training set held out of gradient updates).

### 2.4 Overfitting analysis

The training dynamics show no evidence of overfitting at the selected checkpoint:

**Negligible train-eval gap at epoch 2.** The gap between training-subset F1 (99.80%) and eval F1 (99.40%) is +0.40 percentage points — well within the range attributable to natural sample heterogeneity between splits of the same corpus.

**Strong eval-to-test transfer.** The held-out test set was never seen during training or model selection, yet test F1 (99.26%) is only 0.14 percentage points below eval F1 (99.40%). A model that had memorised training examples rather than learned patterns would show a substantially larger drop at this point.

**Near-perfect test ROC-AUC (99.97%).** AUC measures ranking quality across all decision thresholds and is sensitive to miscalibrated confidence. A memorising model assigns high confidence to training examples but uncertain scores to unseen data, which would depress AUC markedly. The near-perfect AUC on 2,022 unseen samples rules this out.

**Epoch 3 spike is a training oscillation, not progressive overfitting.** The eval loss rose sharply at epoch 3 (0.0324 → 0.1868) then recovered at epoch 4 (0.0698). This is a known pattern under cosine annealing when the model is near its optimal point — the learning rate briefly destabilises the loss surface before converging again. It is not a sign of accumulating memorisation: the model's MCC recovered from 0.944 to 0.983 at epoch 4 without additional training data.

**Training loss reaching near-zero is expected, not alarming.** By epoch 2, individual training steps report losses of 0.0001. This means the model correctly separates nearly all training examples. The eval loss at the same checkpoint (0.0324) is in the same order of magnitude, not orders apart — the ratio of ~5× is normal for a model that fits training data well without degrading generalisation.

## 3. Results

| Metric | Human (label 0) | Machine (label 1) | Overall |
|---|---|---|---|
| Accuracy | — | — | 99.26% |
| F1 | 99.26% | 99.26% | 99.26% |
| Precision | 99.80% | 98.73% | — |
| Recall | 98.72% | 99.80% | — |
| ROC-AUC | — | — | 99.97% |
| MCC | — | — | 0.9852 |

Table 4. Test set performance (n = 2,022).

| | Predicted: Human | Predicted: Machine |
|---|---|---|
| Actual: Human | 1,000 (TN) | 13 (FP) |
| Actual: Machine | 2 (FN) | 1,007 (TP) |

Table 5. Confusion matrix — test set. 13 false positives (human labelled machine); 2 false negatives (machine labelled human).

The model misclassifies 15 of 2,022 test samples. The asymmetry — 13 false positives versus 2 false negatives — indicates the model is slightly conservative about calling a text machine-generated, which is the safer error direction when falsely accusing a human author carries greater cost. The MCC of 0.9852 accounts for all four cells of the confusion matrix and confirms the result is not an artefact of class balance.

## 4. Data Integrity Audit

A five-part audit was run after training to confirm that the result is not inflated by data artefacts. All checks used only text content and were computed independently of the model weights.

| Audit | Finding | Implication |
|---|---|---|
| **Exact duplicates** (MD5 fingerprint) | 17 human texts in train also appear in eval (1.71%); 18 in test (1.78%). Zero machine duplicates across any split pair. | Upper-bound inflation if all 35 leaked samples were memorised: test accuracy drops from 99.26% to 98.57%. Actual inflation is smaller. |
| **Near-duplicates** (MinHash LSH, 5 thresholds) | At Jaccard ≥ 0.9 (tightest): 76 train-test pairs (3.8% of test). All intra-split near-dups are same-label at every threshold. | Overlap drops sharply as threshold rises; no cross-class content contamination within any split. |
| **Cross-class similarity** | At Jaccard ≥ 0.5: 1 human-machine pair in train (0.01%). Zero in eval or test. Zero at all thresholds ≥ 0.6. | Human and machine texts are near-orthogonal at character level. Classification cannot rely on content overlap between classes. |
| **Vocabulary dominance** | 200 class-dominant words identified in train. 97% consistent in eval, 98% in test. Within-class word-rate stability: 46–63%. | Vocabulary associations are real and stable — a genuine learnable signal, not a spurious shortcut confined to training data. |
| **Label quality** (Cleanlab) | Train: 0 flagged (0.00%). Eval: 5 flagged (0.25%). Test: 9 flagged (0.45%). All flagged samples are human texts. | Labelling is essentially clean. Mean model confidence on all correctly classified samples exceeds 99.2%. |

Table 6. Five-part data integrity audit summary.

## 5. Conclusion

Fine-tuning AfroXLMR-Large-29L on 16,377 balanced isiZulu passages produces a classifier that achieves 99.26% accuracy, 99.97% ROC-AUC, and an MCC of 0.9852 on a fully held-out test set. The train-to-eval F1 gap of 0.40 percentage points and the eval-to-test gap of 0.14 percentage points together confirm the model generalises cleanly to unseen data. The data integrity audit rules out exact leakage, near-duplicate inflation, cross-class content shortcuts, vocabulary memorisation, and label noise as alternative explanations. The result reflects genuine detection of the linguistic differences introduced by back-translation — differences that AfroXLMR, pre-trained specifically on African language corpora, is well-positioned to capture.